

Institute of Actuaries of India

Subject CT4 – Models

May 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1 :

(i) At the end of five deliveries bowled by each bowler the current score is leveled at 1-1 i.e. a 'Tie'.

Australian bowler goes next and if he strikes the stumps then Australia has an advantage. On the other hand if the Australian bowler fails to strike the stumps Ireland has an advantage.

With Australia having an advantage, if the bowler from Ireland strikes the stumps on his attempt the scores are level again i.e. a 'Tie'. On the other hand if the Ireland bowler fails to strike the stumps Australia will win the game.

With Ireland having an advantage (after Australian bowler has failed to hit the stumps on his strike), if the bowler from Ireland strikes the stumps on his attempt Ireland will win the game. On the other hand if the Ireland bowler fails to strike the stumps the scores continue to be on level i.e. a 'Tie'.

Since the probability of moving to the next state does not depend on the history prior to entering the state, Markov property holds.

The state space is defined as follows:

State	Description
L	Scores Level/Tie
A_A	Advantage Australia
A_I	Advantage Ireland
W	Game Won

Alternative solution: It is possible to construct a Markov chain where the terminal state actually indicates who won the game i.e. Australia Wins/Ireland Wins. Full marks should be awarded to such alternative solution.

State space for such a solution would be:

State	Description
L	Scores Level/Tie
A_A	Advantage Australia
A_I	Advantage Ireland
W_A	Australia Wins
W_I	Ireland Wins

[5]

(ii) The transition matrix is:

	L	A _A	A _I	W
L	0	0.9	0.1	0
A _A	0.55	0	0	0.45
A _I	0.45	0	0	0.55
W	0	0	0	1

For the 5-State space the transition matrix is:

	L	A _A	A _I	W _A	W _I
L	0	0.9	0.1	0	0
A _A	0.55	0	0	0.45	0
A _I	0.45	0	0	0	0.55
W _A	0	0	0	1	0
W _I	0	0	0	0	1

[2]

(iii) Given that each of the solutions above has an absorbing state one cannot get from that state to any other, so chain is not irreducible.

[1]

(iv) The absorbing state(s) W (or W_A and W_I) have no period and the other three states L, A_A and A_I have a period of 2.

[1]

(v) After two strikes (one from bowler of each team) from the tie, the tie-breaker would either be completed or be back to tie again.

The probability of returning to tie after two points is given by:

Probability of Australia bowler striking the stumps on his attempt x Probability of Ireland bowler striking the stumps on his attempt

+

Probability of Australia bowler failing to strike the stumps on his attempt x Probability of Ireland bowler failing to strike the stumps on his attempt

$$= 0.9 \times 0.55 + 0.1 \times 0.45$$

$$= 0.54$$

We need to find number of such cycles, N, of returning to tie such that

$$0.54^N = 1 - 95\%$$

Solving the above equation:

$$N = \frac{\ln(0.05)}{\ln(0.54)} = 4.86$$

Since the game can finish in cycles of two strikes, the required number of cycles is 5 i.e. 10 strikes.

[3]

(vi) After two strikes:

a) Australia wins the game (Probability is: $0.9 \times 0.45 = 0.405$)

Or

b) Ireland wins the game (Probability is: $0.1 \times 0.55 = 0.055$)

Or

c) Scores are at level again (Probability is 0.54)

Let W_A and W_I be the probabilities that Australia and Ireland wins the game that is current on level score.

Therefore

$$W_A = 0.405 + 0.54 \cdot W_A$$

Solving for W_A gives:

$$W_A = 0.8804$$

Probability that Ireland eventually wins the game is 0.1196 ($1 - W_A$). Alternatively, this can be determined using the equation:

$$W_I = 0.055 + 0.54 \cdot W_I$$

[4]

Probability of Ireland bowler striking the stumps is 0.55. However, in order for Ireland to win the game, it is conditional that the Australian bowler fails to strike the stumps followed by their bowler striking the stumps. The conditional probability is higher than the overall probability – which is one would reasonably expect

[Total Marks-16]

Solution 2 :

i) A Poisson process with rate λ is a continuous – time integer – valued process $N_t, t \geq 0$ with the following properties ;

$$- N_0 = 0$$

- N_t has independent increments

- N_t has Poisson distributed stationary increments:

$$P[N_t - N_s = n] = [\lambda(t-s)]^n e^{-\lambda(t-s)} / n!, \quad s < t, n = 0, 1, \dots$$

Alternatively

The standard time-homogeneous Poisson process is a counting process in continuous time, $\{N_t, t \geq 0\}$, where N_t records the number of occurrences of some type of event within the time interval from 0 to t . The events of interest occur singly and may occur at any time.

Formally, an integer-valued process $\{N_t, t \geq 0\}$, with filtration $\{F_t, t \geq 0\}$, is a Poisson process if:

$$\begin{aligned} P[N_{t+h} - N_t = 1 | F_t] &= \lambda h + o(h) \\ P[N_{t+h} - N_t = 0 | F_t] &= 1 - \lambda h + o(h) \\ P[N_{t+h} - N_t \neq 0, 1 | F_t] &= o(h) \end{aligned}$$

where the statement that $f(h) = o(h)$ as $h \rightarrow 0$ means $\lim_{h \rightarrow 0} \left(\frac{f(h)}{h} \right) = 0$

Alternatively

The holding times, $T_0, T_1, \dots$ of $\{N_t\}_{t \geq 0}$ are independent exponential random variables with parameter λ and $N_{T_0+T_1+\dots+T_{n-1}} = n$.

Alternatively

$\{N_t\}_{t \geq 0}$ is a Markov jump process with independent increments and transition rates given by:

$$\mu_{ij} = \begin{cases} -\lambda & \text{If } j = i \\ \lambda & \text{If } j = i + 1 \\ 0 & \text{otherwise} \end{cases}$$

ii) As per the independent and stationary increments property of the Poisson process:

$$\begin{aligned}
 \Pr(N_5 = 8; N_9 = 12) &= \Pr(N_5 = 8; N_9 - N_5 = 4) \\
 &= \Pr(N_5 = 8) \Pr(N_9 - N_5 = 4) \\
 &= \frac{(12.5)^8 \exp(-12.5)}{8!} \frac{(10)^4 \exp(-10)}{4!} \\
 &= \frac{12.5^8 10^4 \exp(-22.5)}{8! 4!} \\
 &= 0.001042
 \end{aligned}$$

[3]

iii) As per the independent, stationary increments property of Poisson process:

$$\Pr(N_{k1} = r \mid N_{k2} = n) = \frac{\Pr(N_{k1} = r, N_{k2} = n)}{\Pr(N_{k2} = n)}$$

As in the previous problem:

$$\Pr(N_{k2} = n) = (\lambda * k_2)^n \exp(-\lambda * k_2) / n!$$

$$\Pr(N_{k1} = r; N_{k2} = n) = \Pr(N_{k1}=r ; N_{k2}-N_{k1}=n-r)$$

$$\begin{aligned}
 &= \Pr(N_{k1}=r) \Pr(N_{k2}-N_{k1}=n-r) \\
 &= \frac{(\lambda * k_1)^r \exp(-\lambda * k_1)}{r!} \frac{(\lambda * (k_2 - k_1))^{n-r} \exp(-\lambda * (k_2 - k_1))}{n-r!} \\
 &= \frac{\lambda^n (k_1)^r (k_2 - k_1)^{n-r} \exp(-\lambda k_2)}{r! n-r!}
 \end{aligned}$$

$$\begin{aligned}
 \Pr(N_{k1} = r \mid N_{k2} = n) &= \frac{n!}{r! n-r!} \frac{(k_1)^r (k_2 - k_1)^{n-r}}{(k_2)^n} \\
 &= \frac{n!}{r! n-r!} \frac{(k_1)^r (k_2 - k_1)^{n-r}}{(k_2)^r (k_2)^{n-r}} \\
 &= \frac{n!}{r! n-r!} \frac{(k_1)^r}{(k_2)^r} \frac{(k_2 - k_1)^{n-r}}{(k_2)^{n-r}} \\
 &= \binom{n}{r} p^r (1-p)^{n-r}, \text{ where } p = k_1/k_2
 \end{aligned}$$

[4]

$$\begin{aligned}
 \text{iv)} \quad E(Y_t) &= E(E(\sum_{k=1}^{N_t} X_k / N_t)) \\
 &= E(E(\sum_{k=1}^{N_t} E(X_k) / N_t)) \\
 &= E(mN_t) \\
 &= m\lambda
 \end{aligned}$$

$\text{Var}(Y_t) = \text{Var}(E(Y_t|N_t)) + E(\text{Var}(Y_t | N_t))$ as X_k 's are IID variables

Since, $\text{Var}(Y_t | N_t=n) = n S^2$

So,

$$\text{Var}(Y_t) = \text{Var}(N_t m) + E(N_t S^2) = m^2 \text{Var}(N_t) + S^2 E(N_t) = \lambda(m^2 + S^2)$$

[3]

[Total Marks-13]

Solution 3 :

- i) The probability that a Ship positioned at Port goes to a Wreckage site when it leaves the port is:

$$0.0015 / (0.0015 + 0.4485 + 0.05) = 0.003$$

[1]

- ii) The expected holding time for a Ship in the Repairs and Maintenance State is

$$1/(0.649+0.001+0.15) = 1.25 \text{ months}$$

[1]

- iii) Define: t_i as the expected future lifetime given that the life is in status i presently, which means t_p is expected future life time of ship at Port, t_{RM} is expected future life time of ship in Repairs & Maintenance and t_T is expected future life time of ship in Transit.

We have:

$$t_T = \frac{1}{0.165+0.025+0.01} + \frac{0.025}{0.165+0.025+0.01} t_{RM} + \frac{0.165}{0.165+0.025+0.01} t_p$$

$$t_T = 5 + 0.125 t_{RM} + 0.825 t_p \quad \text{----- (1)}$$

Similarly,

$$t_p = 2 + 0.1 t_{RM} + 0.897 t_T \quad \text{----- (2)}$$

$$t_{RM} = 1.25 + 0.81125t_p + 0.1875 t_T \text{ ----- (3)}$$

Solving the above equations for t_T and t_p gives

$t_T = 136.94$ months, $t_p = 138.79$ months and $t_{RM} = 139.52$ months.

[6]

[Total Marks-8]

Solution 4 :

i) A good model will:

- be valid, rigorous and well documented
- reflect the risk profile of the business being modeled
- allow for all the significant features of the business being modeled
- have appropriate input parameters and parameter values, taking into account any
- special features of the provider and the economic & the business environment
- easy to communicate
- should exhibit sensible joint behavior of model variables
- capable of independent output verification
- not be overly complex
- not be time consuming to run
- be capable of development and refinement
- be capable of being implemented in a range of ways

[4]

ii) The key steps in a modeling process can be described as follows:

- The objective here is to build a model to determine payouts above INR 1mn and thereby arriving at the distribution of large payouts of the betting firm. The distribution of number of payouts above INR 1mn each year and distribution of payout amounts above INR 1mn i.e. model frequency and severity of large payouts.
- To model the payouts, large payouts data needs to be collected. The payout threshold can be selected at 75% (or any suitable value but not less than 50%) of INR 1mn.
- Data could also be sourced from another betting firm in the same or another region or country, reinsurers or from industry body if it exists.
- Assumptions pertaining to increase in Payouts over the years on account of inflation need to be made. This can be determined based on the inputs from the betting house or suitable

RPI/CPI index can be used. If there is no change in the payout ratio on account of inflation no adjustments to the data is needed.

- To model the frequency of large payouts, the payout data needs to be grouped into payouts bands. A suitable band size is selected based on experience or data. Bands with small number of payouts (e.g. less than 5) needs to be combined so as to have credible data. A histogram is determined based on the grouped data. The parameter for the Poisson (or Negative Binomial) distribution is determined based on the data. The goodness of fit of the distribution is validated using a suitable method (e.g. Chi-Square Test).
- To model the severity of large payouts, various distributions (e.g. Lognormal, Pareto etc) are fitted to the payout. The parameters of the distribution is determined using the Method of Moments or MLE method. The goodness of fit of the distribution is validated using a suitable method.
- Stochastic simulation methods are used to generate payouts based on the selected frequency and severity distributions.
- Review and carefully consider the appropriateness of the model in the light of small changes in input parameters.
- The overall output from the model is tested for reasonableness.
- Ensure that any relevant professional guidance has been complied with.
- Communicate and document the results and the model.

A deterministic model does not contain any random components and its output is determined based on fixed inputs and defined relationships between inputs and outputs.

A stochastic model on the other hand recognizes the random nature of the input components and hence for any given values of the inputs the outputs are an estimate of the characteristics of the model. Several independent iterations of the model are required for each set of inputs to study their implications. The output of a stochastic model gives the distribution of relevant results for a distribution of scenarios.

Given that we would like to determine the overall distribution of payouts above INR 1mn a stochastic model is better suited for this purpose.

[8]

[Total Marks-12]

Solution 5 :**i)**

Graduation refers to the process of using statistical techniques to improve the estimated mortality rates provided by the crude rates.

The aims of graduation are to produce

- A smooth set of rates as we believe that the mortality rates are a smooth function of age on basis of large investigation.
- Remove random sampling errors.
- Use the information available from adjacent ages to improve the reliability of the estimates.
- Customers expect premiums to progress smoothly with age.

[3]**ii)**

The advantages and disadvantages of use of parametric formula for graduation are given below:

Advantages:

- The graduated rates will progress smoothly provided the number of parameters is small
- The approach is very well-suited to the production of standard tables from large amounts of data.
- The method can produce extremely precise results
- Can fit the same formula to different experiences and compare parameter values to highlight differences between them.

Disadvantages:

- It can be hard to find a formula to fit well at all ages without having lots of parameters
- Care is required when extrapolating: the fit is bound to be best at ages where we have lots of data, and can often be poor at extreme ages.

[2]**[Total Marks-5]****Solution 6 :**

i)

Tj	Nj	Dj	Dj/Nj	1-(Dj/Nj)
6	20	3	0.1500	0.8500
7	16	1	0.0625	0.9375
10	14	1	0.0714	0.9286
13	11	1	0.0909	0.9091
16	10	1	0.1000	0.9000
22	6	1	0.1667	0.8333
23	5	1	0.2000	0.8000

Then the Kaplan-Meier estimate of the survival function is:

$0 \leq t < 6$	1
$6 \leq t < 7$	0.8500
$7 \leq t < 10$	0.7969
$10 \leq t < 13$	0.7400
$13 \leq t < 16$	0.6727
$16 \leq t < 22$	0.6055
$22 \leq t < 23$	0.5045
$23 \leq t < 34$	0.4036

[4]

ii) An approximate 95% confidence interval for S(8) is:

$$S(8) \pm 1.96 \text{Var} (S(8))^{(1/2)}$$

$$S(8) = 0.7969$$

Using Greenwood's formula:

$$\begin{aligned} \text{Var} (S(8)) &= (0.7969)^2 * [3/340 + 1/240] \\ &= 0.6350 * 0.013 \\ &= 0.0083 \end{aligned}$$

So confidence interval is:

$$0.7969 \pm 1.96 * 0.0083^{(1/2)} = 0.7969 \pm 0.1786$$

$$(0.6183, 0.9755)$$

[3]

iii) Type of censoring present:

Right censoring – cuts short observations that are in progress (here due to the end of the investigation period)

Type I censoring – Because censoring times are known in advance, this is a form of Type I censoring

[2]

[Total Marks-9]

Solution 7 :

i) The major reasons company would like to perform mortality investigation are:

- In order to determine the assumptions for pricing of the new products or pricing of the existing products.
- To determine the basis for statutory valuations or embedded value
- To see the trend in the mortality experience of the company including improvement in mortality
- To analyse the major causes of death for the company
- To determine any anti-selection
- To review the underwriting limits for the products
- Splitting the mortality experience by factors will give management information about causes of risk for the company rising from mortality
- To fulfil statutory reporting requirements

[3]

ii) The calculations are provided in the table below:

Age	Initial exposed to risk	Standard Mortality rates (IALM 94- 96)	Standard Mortality rates (IALM 06- 08)	Actual reported deaths at date of investigati on	Actual deaths (including IBNR - rounded up)	Expected deaths (IALM 94- 96)	Expected deaths (IALM 06- 08)	$(A'-E1)^2$ /E1	$(A'-E2)^2$ /E2
x	Ex	qx(1)	qx(2)	A	A'	E1	E2	Chi1	Chi2
40	1,00,000	0.002053	0.001803	130	169	205.3	180.3	6.42	0.71
41	98,927	0.002247	0.001959	142	185	222.3	193.8	6.26	0.40
42	98,100	0.002418	0.002140	155	202	237.2	209.9	5.23	0.30
43	97,450	0.002602	0.002350	170	221	253.6	229.0	4.18	0.28
44	96,600	0.002832	0.002593	185	241	273.6	250.5	3.88	0.36
45	95,800	0.003110	0.002874	205	267	297.9	275.3	3.21	0.25
Total				987	1,285	1,490	1,339	29.17	2.30

H0: Standard mortality table IALM 94-96 represents the true underlying mortality rates of the experience

The degrees of freedom = 5

Observed value chi 1 = 29.17

Tabular value of chi square for upper 95% point = 11.1

The observed value of test statistic is more than 5% significance point

Hence, we reject the null hypothesis

H0: Standard mortality table IALM 06-08 represents the true underlying mortality rates of the experience

The degrees of freedom = 5

Observed value chi 1 = 2.3

Tabular value of chi square for upper 95% point = 11.1

The observed value of test statistic is less than 5% significance point

Hence, there is no evidence to reject the null hypothesis

Conclude that IALM 06-08 appears to be a better fit than IALM 94-96 standard table

[5]

- iii) Chi square test is based on squared deviations; it tells us nothing about the direction of any bias.

In the calculation above the own experience is consistently lower than both the standard tables particularly the IALM 94-96 table.

[1]

- iv) Under the null hypothesis the number of positive signs should have a Binomial (6, 0.5) distribution.

The probability of obtaining 0 positive signs if the null hypothesis is true is 0.015

Since this is less than 0.025 (two-tailed test), the experience is systematically lower (or higher) than those derived from the IALM 06-08 tables.

[2]

- v) While looking at the A-E1 values it is quite evident that our experience is consistently lower than the IALM 94-96 table. This is due to the improvement in mortality which has taken place as IALM 06-08 is based on more recent data.

Also we have observed that the actual experience is lower than IALM 06-08 albeit to a lower extent compared to IALM 94-96 tables. We can infer that mortality improvement has resulted in lower experience for CY 2013 as IALM 06-08 is not based on the most recent data.

[2]

[Total Marks-13]

Solution 8 :

- i) The principle of correspondence states that a life alive at time t should be included in the exposure at age x at time t if and only if were that life to die immediately, he or she would be counted in the deaths data dx at age x .

[1]

- ii) $P_x(t)$ is the number of policies under observation aged x nearest birthday on 1 January in year t .

Let the number of policies aged x last birthday on 1 January in year t be $P_x'(t)$

Then, assuming that birthdays are evenly distributed,

$$P_x'(t) = 0.5 * [P_x(t) + P_{x+1}(t)]$$

Using the trapezium approximation this is

$$Ex^{\circ} = 0.5 * [P_x'(t) + P_x'(t+1)]$$

substituting for the $P_x(t)$ in terms of $P_x'(t)$ from the equation above produces

$$Ex^{\circ} = [0.5 * 0.5 * [P_x(t) + P_{x+1}(t)] + 0.5 * [P_x(t+1) + P_{x+1}(t+1)]]$$

[4]

[Total Marks-5]

Solution 9 :

The major limitations of performing the mortality analysis by claims are:

- Multiple causes of death might be reported which might make it difficult to classify the policy for a particular cause of death.
- The main underlying cause of death (out of multiple causes) might not be reported.
- The cause of death might not be available or not known especially policies pertaining to rural areas.
- Some causes of deaths might be falsely reported like
 - the actual cause of death is AIDS due to social reasons
 - in case of fraudulent claims
 - suicide due to social reasons
- The cause of claim on the death certificate might be heavily influenced by expertise of the medical practitioners/certifier so incorrect cause of claim might be reported.

[3 Marks]

Solution 10 :

- i) The decreasing Weibull distribution is used to model hazards which decrease with time.
Example for decreasing Weibull distribution could be infant mortality rates.

[1]

- ii) The constant hazard Weibull distribution is the exponential distribution.

$$P(Y > t | X > s) = P(X > s + t | X > s)$$

$$= P(X > s + t, X > s) / P(X > s)$$

$$=P(X > s + t)/P(X > s)$$

$$=e^{-(s+t)}/e^{-(s)}$$

$$= e^{-(t)}$$

[2]

iii) Say after 100 months, the survival probability of the membrane is

$$S(100) = e^{-(100/240)^2}$$

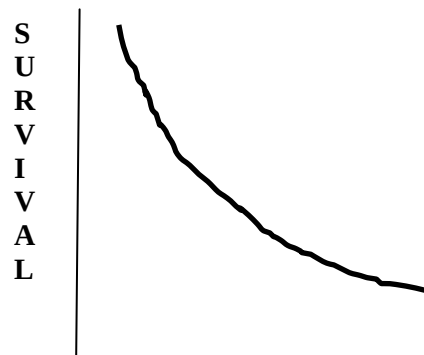
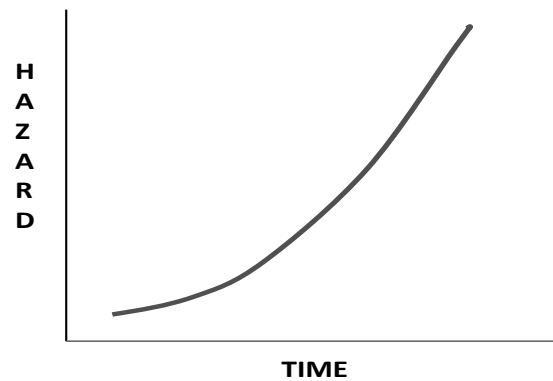
$$= 0.846$$

Say after 200 months, the survival probability of the membrane is

$$S(200) = e^{-(200/240)^2}$$

$$= 0.499$$

Hence the survival probability decreases as time elapses. Weibull distribution where the failure rate is greater than 1 is the increasing hazard distribution.



TIME

50% survival is achieved after 199.81 months.

$$0.5 = e^{-(x/240)^2}$$

$$-\ln(0.5) = (x/240)^2$$

$$X = 240 * (-\ln(0.5))^{(1/2)}$$

$$= 199.81$$

[4]

[Total Marks-7]

Solution 11 :

- i) Maximum-likelihood estimation (MLE) is a method that finds the most likely value of parameter(s) based on the collected data set. Example - assume that the accidents follow Poisson distribution with some unknown mean and variance, the mean and variance can be estimated with MLE while having sample values from the overall population.

[2]

ii)

$$\begin{aligned} \ln L(x_1, x_2, \dots, x_5; \lambda_0) &= \ln f_X(x_i; \lambda_0 \text{ age}_i) \\ &= \sum \ln(e^{-(\lambda_0 \text{ age}_i)} (\lambda_0 \text{ age}_i)^{x_i}) - \ln(x_i!) \\ &= \sum \ln(e^{-(\lambda_0 \text{ age}_i)} (\lambda_0 \text{ age}_i)^{x_i}) - \ln(x_i!) \\ &= \sum -(\lambda_0 \text{ age}_i) + x_i \ln(\lambda_0 \text{ age}_i) - \ln(x_i!) \\ &= -(\lambda_0 \sum \text{age}_i) + \sum x_i \ln(\lambda_0 \text{ age}_i) - \sum \ln(x_i!) \end{aligned}$$

Now let's take the partial with respect to λ_0

$$\begin{aligned}
 \frac{\frac{\partial \ln L(x_1, x_2, \dots, x_5; \lambda_0)}{\sum \ln(x_i!) \partial \lambda_0}}{\partial \lambda_0} &= \frac{-\lambda_0 \sum \text{age}_i + \sum x_i \ln(\lambda_0 \text{age}_i) -}{\partial \lambda_0} \\
 &= \frac{-\sum \text{age}_i + \sum x_i \frac{\partial \ln(\lambda_0 \text{age}_i)}{\partial \lambda_0}}{\partial \lambda_0} \\
 &= \frac{-\sum \text{age}_i + \sum x_i \text{age}_i}{\lambda_0 \text{age}_i} \\
 &= \frac{-\sum \text{age}_i + \sum x_i}{\lambda_0} \\
 &= \frac{-5 \text{ age}^- + 5 x^-}{\lambda_0}
 \end{aligned}$$

Set this equal to zero and solve for λ_0

$$\begin{aligned}
 \lambda_0 \left(-5 \text{ age}^- + \frac{5 x^-}{\lambda_0} \right) &= 0 \\
 \lambda_0 &= x^- / \text{age}^- \\
 \lambda_{\text{MLE}} &= \text{age}_i * x^- / \text{age}^-
 \end{aligned}$$

[5]

iii)

Age of the individual	Number of accidents the individual has had given the age
18	0
25	1
34	2
44	2
80	4

$$\lambda_0 = \bar{x} / \bar{\text{age}}$$

$$\text{Therefore, } \lambda_0 = 0.04478$$

$$P(x = 1, \text{age } 25) = 0.37$$

$$P(x = 1, \text{age } 80) = 0.10$$

Given the sample the probability of just having one life threatening accident at age 80 is lot lower than at age 25.

[2]
[Total Marks-9]
